

# Methods of Applied Mathematics

## Homework 6

Revised coursework manuscript

### 1. Deep Water Waves

It is shown in fluid dynamics that ocean waves in deep water that are propagating in the positive  $x$ -direction are governed by the dispersion relation

$$\omega(k) = \sqrt{gk},$$

where  $k > 0$  is the wavenumber and  $g = 9.81 \text{ m/s}^2$  is gravity.

1. Sketch  $\omega(k)$ .
2. Compute the phase velocity  $c_p$  and the group velocity  $c_g$ , and indicate their geometric meaning in the sketch.
3. What is the wavelength of a wave with period of ten seconds, and what is its group velocity?
4. Suppose a storm occurs at time  $T$  and a distance  $L$  away from the shore, which creates surface waves that fan out in a slowly varying wavetrain and then eventually reach the shore. Show that for deep water waves you would expect the slowly varying frequency  $\omega$  observed at the shoreline to grow linearly in time.
5. If on Monday the observed wave period is 15 seconds and on Tuesday it is 7.5 seconds, what day prior did the storm occur, and how far from the shore was it?

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\*Revision note: the shallow-water group-velocity formula in Problem 2 and the integral-level conjugation argument in Problem 3 have been corrected. The remaining text and figures follow the original coursework manuscript.

## Solution to Problem 1

Sketch  $\omega(k)$

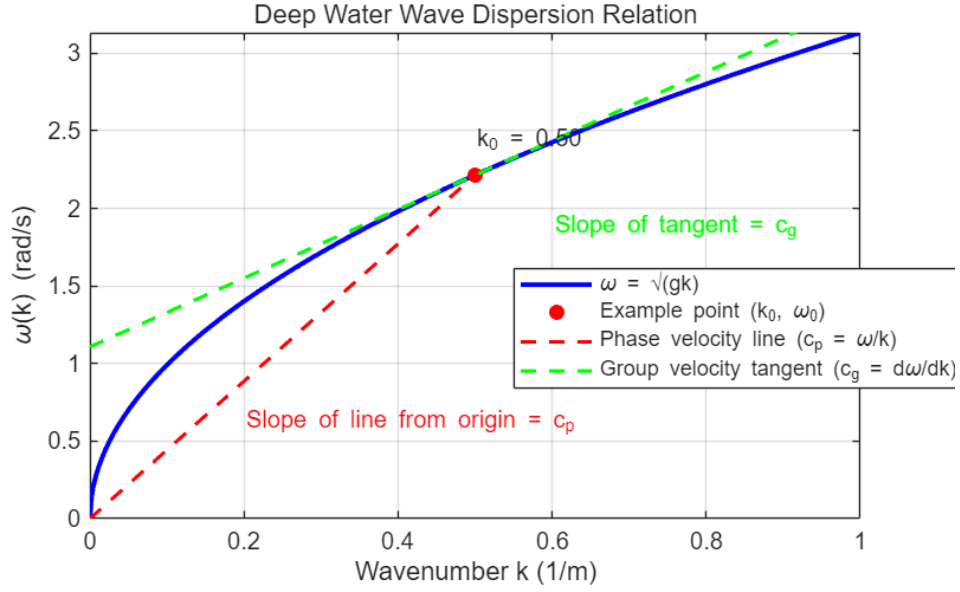


Figure 1:  $\omega(k) = \sqrt{gk}$

Compute  $C_g$  and  $C_p$

$$c_p = \frac{\omega}{k} = \frac{\sqrt{gk}}{k} = \sqrt{\frac{g}{k}}.$$

$$c_g = \frac{d\omega}{dk} = \frac{1}{2}\sqrt{\frac{g}{k}}.$$

Thus,

$$c_g = \frac{1}{2}c_p.$$

Therefore, **the group velocity is always half the phase velocity.**

### Wavelength and its group velocity

Wavelength  $\lambda$  is

$$\lambda = \frac{2\pi}{k},$$

and

$$k = \frac{\omega^2}{g}.$$

Thus, when  $T = 10s$ ,

$$\lambda = \frac{g}{\omega^2}2\pi = \frac{gT^2}{4\pi^2}2\pi = \frac{gT^2}{2\pi} = 156.13m$$

And.

$$c_g = \frac{1}{2}\sqrt{\frac{g}{k}} = \frac{1}{2}\sqrt{\frac{g\lambda}{2\pi}} = \frac{gT}{4\pi} = 7.8m/s$$

$\omega$  grow linearly in time

If we observe at shore,

$$t = T + \frac{L}{c_g(k)} = T + \frac{2L\omega}{g}.$$

Thus,

$$\omega(t) = \frac{g}{2L}(t - T),$$

which shows that for deep water waves grow linearly in time.

### Day of the storm and its distance from the shore

By

$$\omega = \frac{2\pi}{T},$$

$\omega_1 = \frac{2\pi}{15}$  on Monday and  $\omega_2 = \frac{2\pi}{7.5}$  on Tuesday. And

$$\omega_2 - \omega_1 = \frac{g}{2L}(t_2 - t_1),$$

where  $t_2 - t_1 = 1\text{day} = 86400s$  is the time between Monday and Tuesday. Thus,

$$L = \frac{g(t_2 - t_1)}{2(\omega_2 - \omega_1)} = 1.01 \times 10^3 km.$$

Then,

$$t_1 - T = \frac{2L\omega}{g} = \frac{2 \frac{g(t_2 - t_1)}{2(\omega_2 - \omega_1)} \cdot \omega_1}{g} = t_2 - t_1.$$

Therefore,

The day prior is 1 day and the distance L is  $1.01 \times 10^3 km$ .

## 2. Deep and Shallow Water Waves

More generally, water waves in a fluid of depth  $H$  propagating in the positive  $x$ -direction are governed by the dispersion relation

$$\omega(k) = \sqrt{gk \tanh(kH)}, \quad k > 0.$$

1. Sketch  $\omega(k)$  and show that it reduces to  $\omega = \sqrt{gk}$  in the deep-water limit  $kH \gg 1$ .
2. Find the limiting form in the opposite, shallow-water limit  $kH \ll 1$ .
3. Are the waves in this limit dispersive?
4. Consider a Tsunami that has a wavelength of 100 km in an ocean of depth 4 km. Does it behave like a shallow-water wave or a deep-water wave? What is its group velocity?

### Solution to Problem 2

Sketch  $\omega(k)$

Here is the sketch:

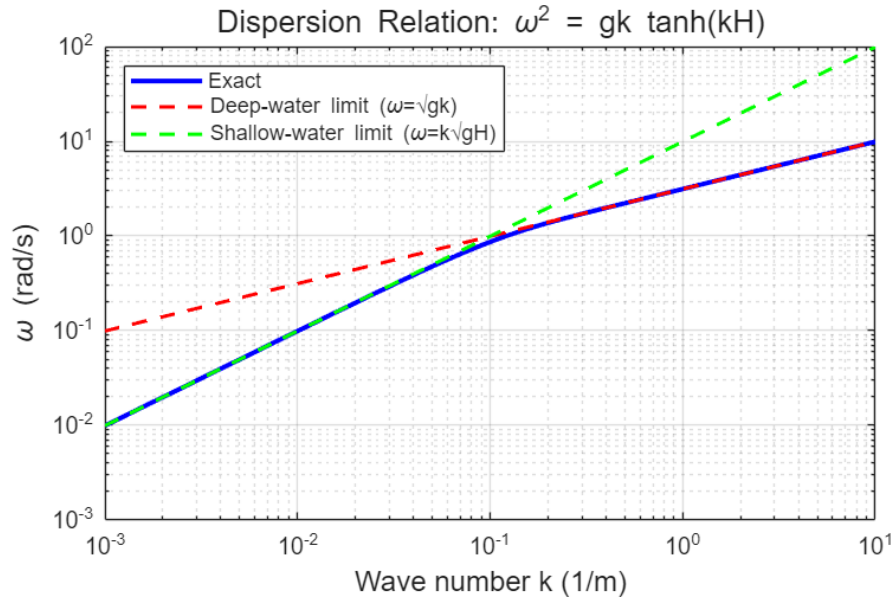


Figure 2:  $\omega(k) = \sqrt{gk \tanh(kH)}$

The definition of

$$\tanh(x) = \frac{e^{2x} - 1}{e^{2x} + 1}.$$

When  $kH \rightarrow \infty$ ,

$$\tanh(kH) \rightarrow 1.$$

Therefore,

$\omega(k) \text{ reduces to } \omega = \sqrt{gk} \text{ in the deep water limit } kH \gg 1.$

## The limiting form of shallow water

When  $kH \ll 1$ ,

$$\tanh kH \approx kH.$$

Thus,

$$\boxed{\omega \approx k\sqrt{gH}.$$

## Is this limit dispersive?

In shallow water,

$$c_p = \frac{\omega}{k} \approx \frac{k\sqrt{gH}}{k} = \sqrt{gH},$$
$$c_g = \frac{d\omega}{dk} = \sqrt{gH}.$$

Thus,

$$c_g = c_p,$$

and they don't change with wavenumber  $k$ . Therefore,

The waves in this shallow water don't change with  $k$ . They are not dispersive!

## Behave like which wave? What is group velocity?

According to the data,

$$k = \frac{2\pi}{\lambda} \Rightarrow kH = \frac{2\pi H}{\lambda} = 8\pi/100 \ll 1.$$

So, it behaves like a **shallow water**.

$$c_g \approx \sqrt{gH} \approx 198.1 \text{ m/s}.$$

### 3. A Stone Dropped in a Pond

For waves in deep water with surface tension, the dispersion relation is

$$\omega^2 = g|k| + \gamma|k|^3 \quad \Rightarrow \quad \omega_{1,2}(k) = \pm\sqrt{g|k| + \gamma|k|^3},$$

where  $g = 981 \text{ cm/s}^2$  and  $\gamma = 74 \text{ cm}^3/\text{s}^2$ .

1. Sketch  $\omega(k)$  for  $k > 0$  and note its asymptotic approximations for very small and very large values of  $k$ .
2. Let  $u(x, t)$  describe the surface elevation with initial conditions

$$u(x, 0) = \delta(x), \quad u_t(x, 0) = 0.$$

Show that

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ikx} \cos(\omega(k)t) dk.$$

3. Using  $\omega(k) = \omega(-k)$ , show that this is equivalent to

$$u(x, t) = \frac{1}{2\pi} \Re \int_{-\infty}^{+\infty} e^{i[kx - \omega(k)t]} dk.$$

4. Consider this integral in the region  $x > 0$  and use the specifics of the dispersion relation to find the particular group velocity ray  $x = ct$  along which the solution decays slowest. Indicate the corresponding wavenumber in your sketch and compute the wavelength that travels along that ray in centimetres.

## Solution to Problem 3

### Sketch $\omega(k)$

Here is the sketch: For deep-water waves, the dispersion relation is

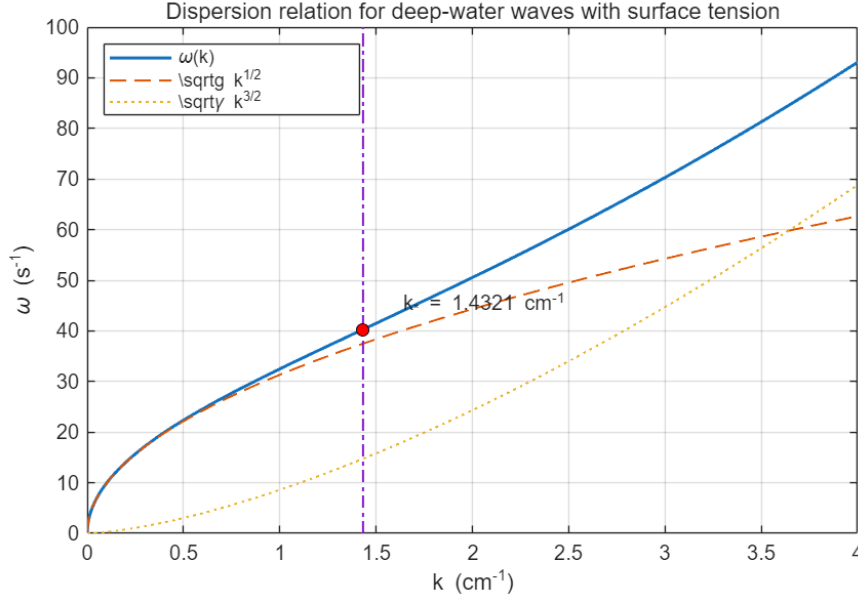


Figure 3:  $\omega^2 = g|k| + \gamma|k|^3$

$$\omega(k) = \sqrt{gk + \gamma k^3}, \quad g = 981 \text{ cm/s}^2, \quad \gamma = 74 \text{ cm}^3/\text{s}^2.$$

- **Long-wave limit, ( $k \ll 1$ ):**

$$\omega(k) \sim \sqrt{gk} = \sqrt{g} k^{1/2}, \quad \sqrt{g} = 31.32 \text{ cm}^{1/2}/\text{s}.$$

Hence,  $\omega \propto k^{1/2}$ .

- **Short-wave limit,  $k \gg 1$ :**

$$\omega(k) \sim \sqrt{\gamma} k^{3/2}, \quad \sqrt{\gamma} = 8.60 \text{ cm}^{3/2}/\text{s}.$$

Thus,  $\omega \propto k^{3/2}$  for large  $k$ .

### Solution via Fourier Transform

Applying Fourier transform,

$$\hat{u}(k, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ikx} dx,$$

each mode satisfies:

$$\hat{u}_{tt}(k, t) + \omega(k)^2 \hat{u}(k, t) = 0.$$

Thus, the general solution is

$$\hat{u}(k, t) = A(k) \cos[\omega(k)t] + B(k) \sin[\omega(k)t].$$

Using the initial conditions  $u(x, 0) = \delta(x)$  and  $u_t(x, 0) = 0$ , it gives  $\hat{u}(k, 0) = 1$  and  $\hat{u} * t(k, 0) = 0$ . Hence,  $A(k) = 1, B(k) = 0$ . Therefore,

$$\hat{u}(k, t) = \cos[\omega(k)t],$$

and by the inverse Fourier transform,

$$u(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} \cos[\omega(k)t] dk.$$

**Show the equivalence**

$$\begin{aligned} u(x, t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} \cos(\omega(k)t) dk \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} \frac{e^{i\omega(k)t} + e^{-i\omega(k)t}}{2} dk \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{2} (e^{ikx+i\omega t} + e^{ikx-i\omega t}) dk \end{aligned}$$

Define the two oscillatory integrals

$$I_{\pm} = \int_{-\infty}^{+\infty} e^{i[kx \pm \omega(k)t]} dk.$$

The preceding expression is  $u = (I_+ + I_-)/(4\pi)$ . Since  $\omega$  is even, complex conjugation followed by the change of variable  $q = -k$  gives

$$\overline{I_-} = \int_{-\infty}^{+\infty} e^{-ikx+i\omega(k)t} dk = \int_{-\infty}^{+\infty} e^{iqx+i\omega(q)t} dq = I_+.$$

Thus  $I_+ + I_- = 2\Re I_-$ , establishing the identity at the integral level. Therefore,

$$u(x, t) = \frac{1}{2\pi} \Re \int_{-\infty}^{+\infty} e^{i[kx - \omega(k)t]} dk.$$

**Compute the wavelength**

Consider the phase  $\phi(k) = kx - \omega(k)t$ . Stationary points satisfy

$$\phi'(k) = 0 \quad \Rightarrow \quad \frac{x}{t} = \omega'(k),$$

where  $\omega'(k)$  is the group velocity. The wave decays most slowly along the ray where the stationary point is degenerate, i.e.

$$\omega''(k) = 0.$$

For  $\omega(k) = \sqrt{gk + \gamma k^3}$ ,

$$\omega'(k) = \frac{g + 3\gamma k^2}{2\sqrt{gk + \gamma k^3}},$$

and solving  $\omega''(k) = 0$  gives

$$12\gamma k(gk + \gamma k^3) - (g + 3\gamma k^2)^2 = 0,$$

$$k_* \approx 1.432 \text{ cm}^{-1}, \quad c_* = \omega'(k_*) \approx 17.83 \text{ cm/s}, \quad \lambda_* = \frac{2\pi}{k_*} \approx 4.39 \text{ cm}.$$

Thus, the slowest-decaying ray is  $x = c_* t$ , corresponding to a wavelength of approximately 4.39 cm.

Here is the sketch:

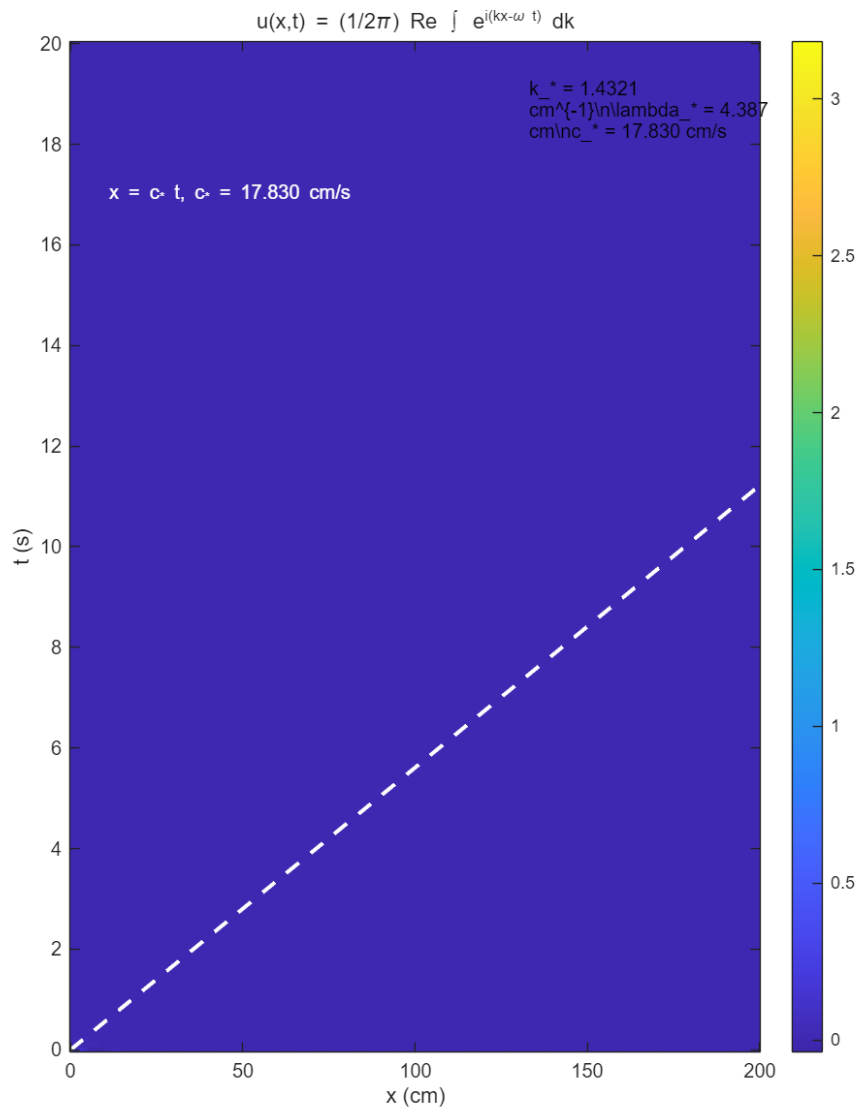


Figure 4: Decaying ray and wavenumber